

# Physics Worksheet: Capacitor Charge & Discharge

## Detailed Answer Key (Explanations)

Square Signal Excitation (LFG) – MCQ Version

### Technical Context: Circuit Parameters

This document provides the correct choices and detailed technical justifications for each question on the MCQ worksheet. It is intended for teacher reference or student self-correction.

## Detailed Solutions

1. **C.**  $u_G = u_C + u_R$

*Explanation:* According to the Law of Addition of Voltages (Kirchhoff's Voltage Law) in a series circuit, the total voltage delivered by the generator at any instant  $t$  is equal to the sum of the voltages across the receiving components in the loop.

2. **B.** Constant and equal to  $E$ .

*Explanation:* The technical context explicitly states that during the first half-period ( $0 \leq t \leq T/2$ ), the low-frequency generator delivers a constant positive voltage  $E$ , effectively acting as a DC source for the charging phase.

3. **C.** It decreases from  $E$  towards 0.

*Explanation:* At  $t = 0$ , the uncharged capacitor acts as a short circuit ( $u_C = 0$ ), so the entire generator voltage falls across the resistor ( $u_R = E$ ). As the capacitor charges,  $u_C$  increases, and thus  $u_R = E - u_C$  must decay exponentially toward 0.

4. **C.**  $i = C \frac{du_C}{dt}$

*Explanation:* Current is the rate of flow of charge:  $i = \frac{dq}{dt}$ . Since the charge on a capacitor is  $q = C \cdot u_C$ , taking the derivative with respect to time (where  $C$  is a constant) yields  $i = C \frac{du_C}{dt}$ .

5. **B.**  $u_R = -u_C$

*Explanation:* During the discharging phase, the generator voltage drops to 0 ( $u_G = 0$ ). Applying the additive law of voltages:  $0 = u_C + u_R$ , which rearranges to  $u_R = -u_C$ .

6. **C.** The current is flowing in the opposite direction to the charging phase.

*Explanation:* According to Ohm's Law,  $u_R = R \cdot i$ . Because resistance  $R$  is always positive, a negative  $u_R$  strictly implies a negative current  $i$ . This means the discharge current flows in the opposite direction to the positive reference direction established during charging.

7. **D.** Seconds (s)

*Explanation:* The time constant is defined as  $\tau = R \cdot C$ . A dimensional analysis shows that Ohms  $\times$  Farads yields units of time, specifically seconds in the SI system.

8. **B.** 63%

*Explanation:* The equation for the charging capacitor is  $u_C(t) = E(1 - e^{-t/\tau})$ . Substituting  $t = \tau$  yields  $u_C(\tau) = E(1 - e^{-1}) \approx E(1 - 0.37) = 0.63E$ , meaning it has reached roughly 63% of its maximum charge.

9. **C.  $T \geq 10\tau$**   
*Explanation:* A capacitor is practically fully charged at  $t = 5\tau$ . For this to happen within the first half-period, we need the half-period to be at least  $5\tau$ . Therefore,  $T/2 \geq 5\tau$ , which gives  $T \geq 10\tau$ .
10. **A.  $f_{max} = \frac{1}{10RC}$**   
*Explanation:* Frequency is the reciprocal of the period ( $f = 1/T$ ). Since the condition for complete charging/discharging is  $T \geq 10\tau$ , substituting  $\tau = RC$  gives  $T \geq 10RC$ . Inverting this relationship for frequency gives  $f \leq \frac{1}{10RC}$ . The maximum frequency is the boundary value.
11. **A. (Exponential growth curve to asymptote  $E$ )**  
*Explanation:* Graph A shows the characteristic exponential charging curve. The voltage starts at 0 and asymptotically approaches the maximum generator voltage  $E$ .
12. **B. Finding the abscissa of the intersection point between the tangent at  $t = 0$  and the horizontal asymptote  $u_C = E$ .**  
*Explanation:* Mathematically, the tangent to the curve  $u_C(t)$  at the origin intersects the maximum voltage asymptote ( $u_C = E$ ) at exactly the time constant  $t = \tau$ . This is a standard graphical method for determining  $\tau$ .
13. **A. (Exponential decay curve from  $E$  to 0)**  
*Explanation:* During discharge, the capacitor starts fully charged at  $E$  and exponentially releases its energy, causing the voltage to asymptotically approach 0 without ever crossing into negative values. This matches Graph A.
14. **B. (Exponential decay curve from  $E$  to 0)**  
*Explanation:* As established in question 3,  $u_R$  starts at  $E$  when the current is at its maximum initial surge, and exponentially decays toward 0 as the capacitor pushes back with its own rising voltage.
15. **D. It starts at a maximum negative value  $-\frac{E}{R}$  and approaches zero exponentially.**  
*Explanation:* At the start of discharge,  $u_G = 0$  and  $u_C \approx E$ . Therefore,  $u_R = -E$ . Using  $i = u_R/R$ , the initial current is  $-E/R$ . As the capacitor discharges,  $u_R$  goes to 0, and current approaches 0 from the negative side.
16. **B. Press the "INVERT" (or reverse) button on Channel 2 because it measures  $-u_R$ .**  
*Explanation:* In a standard setup where the oscilloscope's ground is placed between the capacitor and the resistor, Channel 1 measures  $u_C$  correctly relative to ground. However, Channel 2 measures the voltage across the resistor in the opposite direction to the current flow (it measures  $-u_R$ ). Inverting Channel 2 corrects the display to show the true  $u_R$ .

